



INDIAN INSTITUTE OF
TECHNOLOGY, ROPAR

HS202 PROJECT REPORT

Human Geography and Societal Needs

College Medical Assistance Website: **CampusMed+**

Group 11

Group Members: -

- i. Chinmay Mittal
- ii. Dhepe Makarand Vishwanath
- iii. Gautam Thapa
- iv. Girish Kuldeep
- v. Hadiya Pratik Ghanshyambhai
- vi. Kanav Dumra
- vii. Sarthak Mangesh Aher
- viii. Shivam

TABLE OF CONTENTS

- ✓ Abstract
- ✓ Definition of Problem
 - Problem Statement
 - Origin / Identification of Problem
 - Methodology Adopted to identify the Problem
 - Survey Analysis
 - Current Development in the Domain
 - Need and Significance of Resolving the Problem
- ✓ Aims / Objectives / Goals
- ✓ Tools and Techniques (Interventions)
- ✓ Detailed Work Plan / Tech Interventions (Workflow)
- ✓ Novelty / Innovations of the Proposed Intervention
- ✓ Approaches to Implementation
- ✓ Possible Constraints and Barriers
- ✓ Expertise Available with Each Student
- ✓ Expected Outcomes
- ✓ Suggested Plan of Action for Utilization of Outcomes
- ✓ Conclusion
- ✓ Contribution of Each Student

Abstract

Healthcare accessibility and efficient record management are critical components of any institutional ecosystem, especially within residential campuses such as IIT Ropar. Despite the presence of a functioning medical center, the existing system largely relies on manual record-keeping and traditional consultation methods, which significantly limits efficiency, accessibility, and scalability. Students often face barriers such as long waiting times, limited doctor availability, and inconvenience in physically visiting the medical center, leading to delayed treatment and over-reliance on self-medication.

This project proposes the development of a digital medical assistance and record management system tailored specifically for the IIT Ropar medical center. The proposed system aims to streamline healthcare delivery by digitizing patient records, enabling structured appointment management, and improving coordination between students, doctors, and medical staff. Unlike commercial healthcare platforms, this system is designed to function within institutional constraints, particularly addressing concerns of data privacy and restricted cloud usage.

The project is grounded in a need-based approach, where the problem was identified through direct interaction with stakeholders, including medical staff and administrative authorities. Based on these interactions, the project evolves not merely as a technological solution but as a practical model that can be presented to the institute administration for future implementation. The system emphasizes efficiency, data security, and scalability while acknowledging existing infrastructural and manpower constraints.

Definition of the Problem

The central problem addressed in this project is the inefficiency and limitations of the current healthcare management system at the IIT Ropar medical center. The existing setup primarily depends on manual documentation and in-person interactions, which results in poor record maintenance, delays in patient handling, and lack of systematic data tracking. This creates operational inefficiencies for staff and discourages students from seeking timely medical attention.

Problem Statement

The IIT Ropar medical center currently lacks a digital system for managing patient records, consultations, and operational workflows. The dependence on manual processes leads to inefficiency, lack of accessibility, and difficulty in maintaining accurate medical histories.

This results in delayed healthcare delivery, reduced effectiveness of treatment, and overall dissatisfaction among students and medical staff.

Origin / Identification of the Problem

The identification of this problem followed a structured need-based assessment approach involving real-world interaction with stakeholders. Initially, our group visited the medical center at IIT Ropar to understand the ground realities and operational challenges. During this visit, we engaged in discussions with the staff and healthcare professionals to identify existing gaps in the system.

One of the key interactions was with a staff nurse, **Ms. Daljeet Kaur**, who provided a comprehensive overview of the current working system. She explained that the medical center relies entirely on handwritten records for maintaining patient data, prescriptions, and treatment history. According to her, this approach is not only time-consuming but also highly inefficient and prone to errors, especially when handling a large number of patients. She emphasized that maintaining proper records is essential for effective treatment, yet the current system fails to ensure reliability and accessibility of data.

Further insights were gained through a meeting with **Dr. Prabal Banerjee**, Head of the Medical Advisory Committee at IIT Ropar. He highlighted that the institute is already considering transitioning towards a digital system and is in the process of evaluating quotations from external service providers. He also mentioned that certain companies are currently providing similar services in institutions like IIT Kanpur, indicating that such systems are both feasible and beneficial.

Dr. Banerjee encouraged student involvement in developing such a solution, noting that as a technical institute, it would be highly valuable if students contributed to solving real institutional problems. He also pointed out additional requirements such as the need for more medical staff and doctors, as currently only two doctors are available during limited working hours (9 AM to 5 PM), which restricts service availability.

Following this, we consulted **Dr. Shweta Jain** from the Computer Science and Engineering department, who has experience in guiding student-led projects. She acknowledged the feasibility of the project but highlighted a major constraint: due to the sensitive nature of medical data, cloud-based storage may not be permissible. This introduces a significant design consideration, requiring the system to operate within secure, possibly local or restricted network environments.

Through these interactions, the problem was clearly identified as both a technological and infrastructural challenge requiring a carefully designed, privacy-conscious solution.

Methodology Adopted to identify the problem

The methodology followed in this project was qualitative and participatory in nature. It involved direct field visits, stakeholder discussions, and observational analysis. Unlike purely survey-based approaches, this method allowed us to gain deeper insights into operational challenges and institutional constraints.

The process began with on-site observation of the medical center's workflow, followed by informal interviews with staff members to understand daily challenges. These discussions were supplemented by consultations with administrative authorities and faculty members, ensuring that the problem was examined from multiple perspectives.

Survey Analysis

Section A: Basic Information

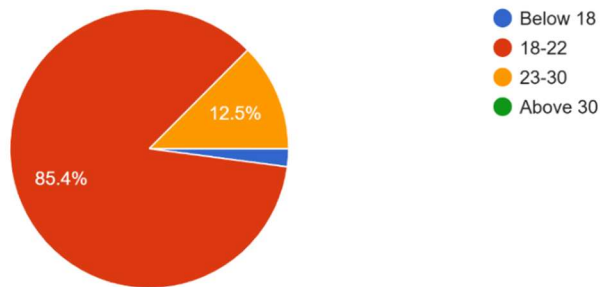
What is your role in the college?

48 responses



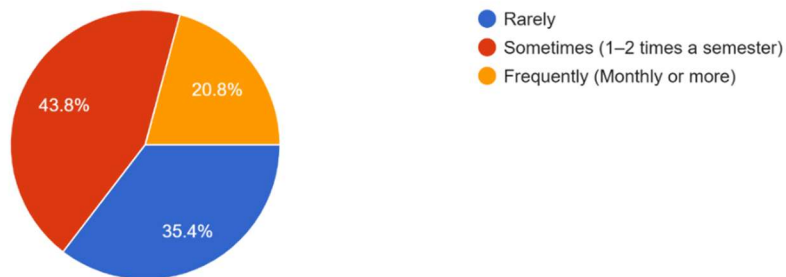
Your age group:

48 responses



How often do you require medical consultation?

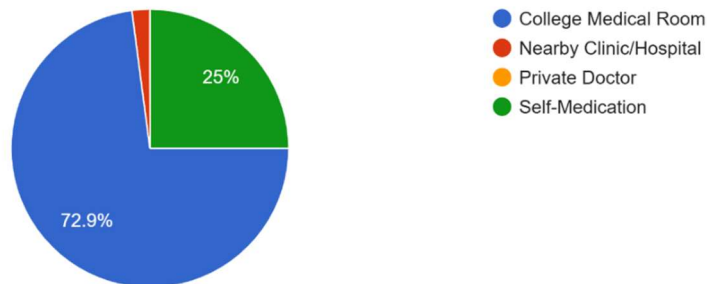
48 responses



Section B: Current Healthcare Access

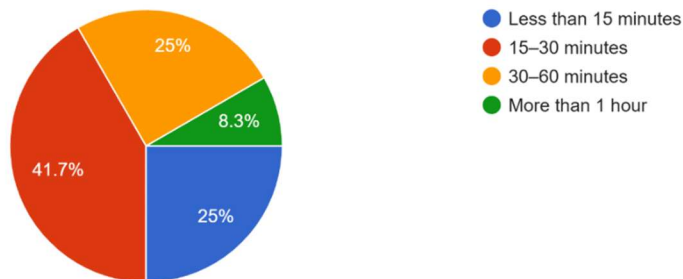
Where do you usually go for medical help?

48 responses



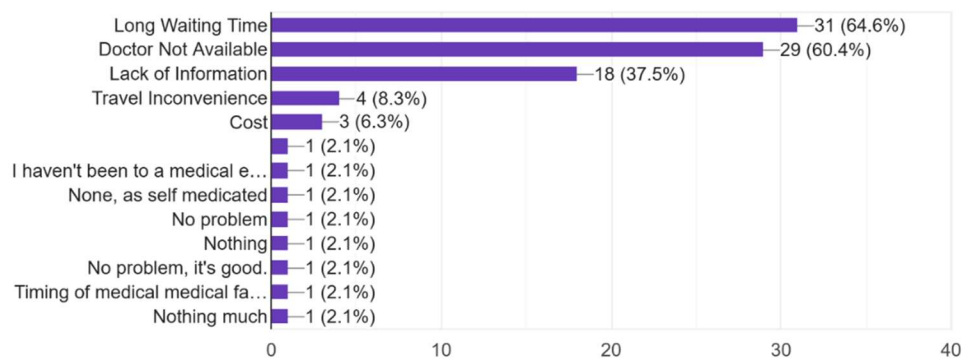
How much time does it usually take to get medical help?

48 responses



What problems do you face while seeking medical help?

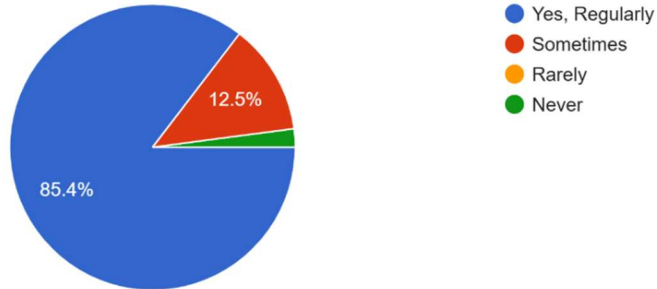
48 responses



Section C: Digital Readiness

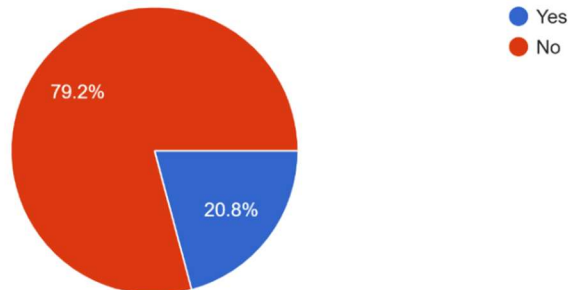
Do you use mobile apps for services like food, transport, or payments?

48 responses



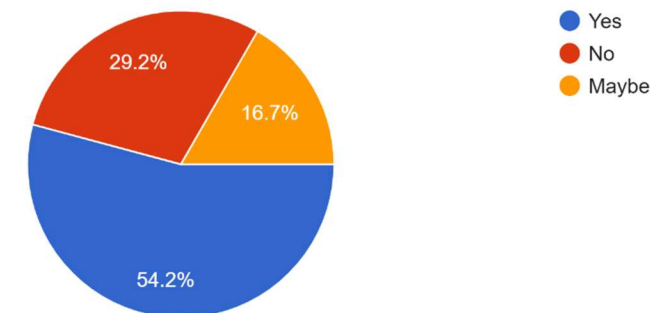
Have you ever used an online consultation app?

48 responses



Are you comfortable sharing health information digitally?

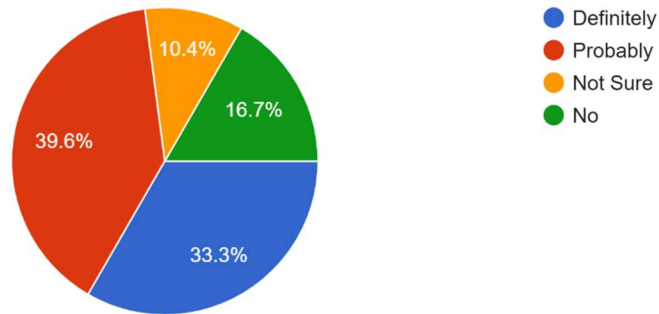
48 responses



Section D: Proposed Healthcare Application

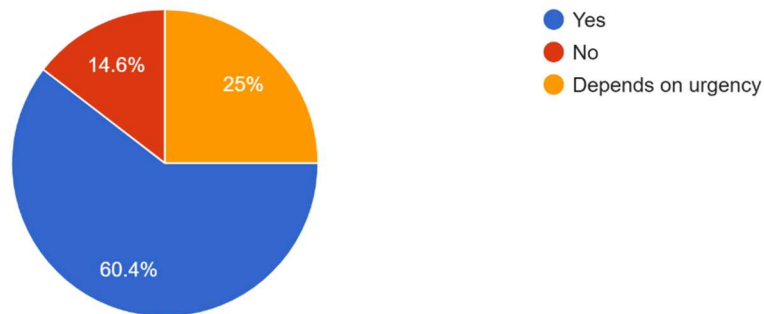
Would you use a college app/website to consult doctors online?

48 responses



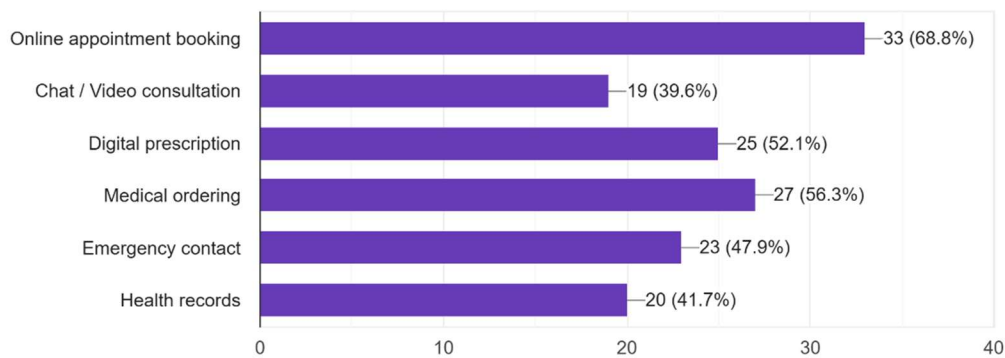
Would you prefer ordering medicines from the college medical shop via app?

48 responses



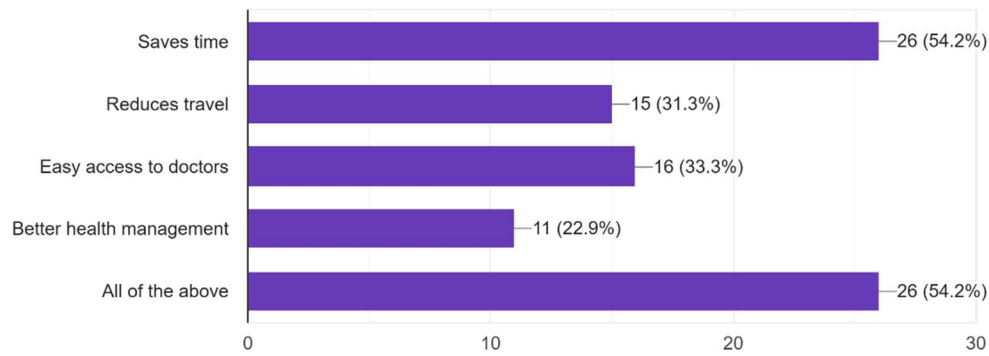
Which features would you find most useful?

48 responses



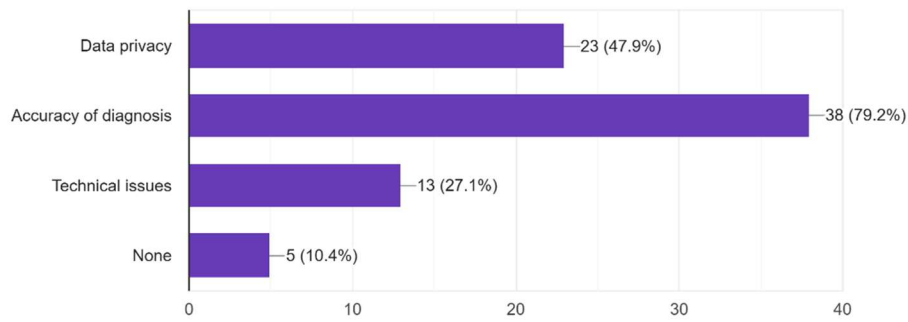
What benefits do you expect from such an app?

48 responses



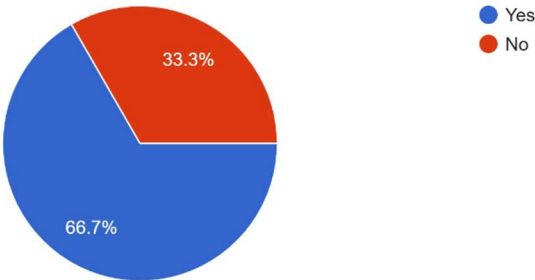
What concerns do you have

48 responses

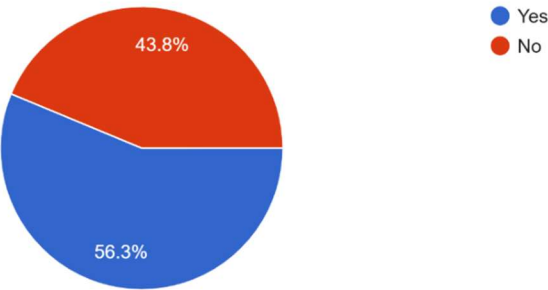


Section E: Healthcare Accessibility & Human Geography Perspective

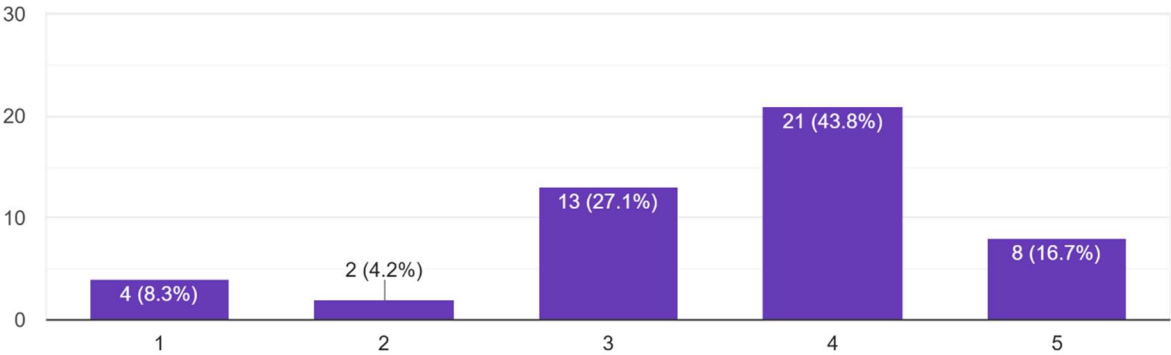
Do you think distance affects your decision to seek medical help?
48 responses



Have you ever delayed treatment due to accessibility issues?
48 responses

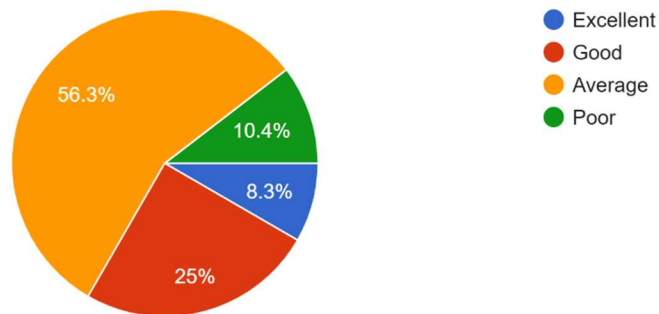


Do you think a digital platform can reduce physical barriers in healthcare?
48 responses



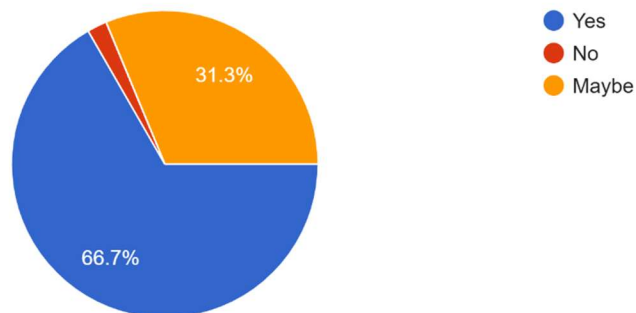
How would you rate current healthcare accessibility in your college?

48 responses



Would this app improve overall well-being of students and staff?

48 responses



The survey conducted among several students, primarily in the 18–22 age group, provides clear insights into healthcare behavior within the campus. It was observed that a significant proportion of students seek medical consultation only occasionally or rarely, indicating a lack of regular engagement with the medical center. While most students prefer visiting the college medical room when required, a considerable fraction still relies on self-medication, highlighting gaps in accessibility and convenience. The major issues reported include long waiting times and limited doctor availability, which act as key deterrents to seeking timely medical attention. At the same time, the survey reveals a high level of digital readiness among students, with the majority actively using mobile applications, although most have not previously used medical apps. Notably, a strong inclination towards adopting a campus-specific digital healthcare solution was observed, with around 72% of respondents expressing interest, thereby reinforcing the feasibility and need for the proposed system.

Current Developments in the Domain

Digital healthcare systems have seen rapid advancements globally, with integrated platforms offering telemedicine, electronic health records (EHR), and automated appointment systems. Many institutions have adopted customized healthcare management systems tailored to their needs.

However, most commercial solutions are designed for large-scale hospitals and often rely on cloud-based infrastructure. While these systems offer convenience, they raise concerns regarding data privacy, especially in institutional settings where sensitive information must be strictly controlled.

Need and Significance of Resolving the Problem

Addressing this problem is crucial for improving healthcare delivery within the campus. A digital system would significantly enhance efficiency, reduce waiting times, and ensure accurate record-keeping. It would also enable better coordination between medical staff and improve overall patient experience.

Moreover, implementing such a system would align with the institute's vision of leveraging technology for practical problem-solving. It would serve as a model for other institutions and demonstrate the potential of student-driven innovation.

Aims / Objectives / Goals

The primary objective of this project is to design a comprehensive and efficient digital healthcare management system that directly addresses the operational inefficiencies currently observed in the IIT Ropar medical center. The project seeks to transform the existing manual processes into a streamlined digital framework that enhances accessibility, reliability, and overall quality of healthcare services within the campus. A major focus is on digitizing patient records in a structured and organized manner so that medical history can be easily stored, retrieved, and utilized for accurate diagnosis and follow-up treatment.

Another key objective is to introduce a systematic appointment scheduling mechanism that reduces overcrowding and waiting time, thereby improving the patient experience. By enabling students to book consultations in advance, the system aims to create a balanced workflow for doctors, ensuring optimal utilization of limited medical resources. Additionally, improving communication between students, doctors, and medical staff is an essential

goal, as it ensures that all stakeholders remain informed and coordinated throughout the treatment process.

A significant objective of this project is to ensure strict compliance with data privacy and security requirements. Given the sensitive nature of medical data, the system is designed to avoid reliance on public cloud infrastructure and instead operate within a secure and controlled environment. This not only protects user data but also aligns with institutional policies and ethical considerations.

Furthermore, the project aims to develop a functional prototype that serves as a proof-of-concept model. This model can be presented to the institute administration to demonstrate feasibility and effectiveness, thereby facilitating future implementation and potential scaling. Ultimately, the goal is to bridge the gap between existing limitations and desired healthcare standards through a practical and sustainable technological intervention.

Tools and Techniques (Interventions)

The proposed solution involves the development of a web-based digital platform that integrates multiple functional components into a single cohesive system. This platform is designed to cater to different categories of users, including students, doctors, and administrative staff, each with their own dedicated interface and functionalities. The use of web-based architecture ensures accessibility across devices while maintaining centralized control over system operations.

At the core of the system lies a robust database management system responsible for securely storing patient records, consultation history, prescriptions, and other relevant medical data. This database is designed to ensure data integrity, consistency, and quick retrieval, which are critical for efficient healthcare delivery. Special attention is given to security mechanisms, including authentication protocols based on institutional credentials, to prevent unauthorized access.

Frontend development tools are utilized to create a user-friendly interface that simplifies navigation and interaction. The design prioritizes clarity and ease of use so that even non-technical users can operate the system without difficulty. On the backend, server-side technologies are employed to manage data processing, system logic, and communication between different components.

In addition to technical tools, the project also incorporates analytical and observational techniques during the problem identification phase. Stakeholder interviews, field visits, and contextual analysis play a crucial role in ensuring that the intervention is aligned with real-

world requirements. This combination of technological and participatory approaches strengthens the overall effectiveness of the proposed solution.

Detailed Work Plan / Technological Interventions (Including Workflow)

The development of this project follows a structured and phased work plan that begins with need assessment and progresses through design, development, and evaluation stages. The initial phase involved visiting the IIT Ropar medical center and engaging with stakeholders to understand existing challenges. These interactions provided valuable insights into system requirements, constraints, and expectations, forming the foundation for subsequent design decisions.

Following the identification of the problem, the system design phase was initiated, focusing on creating a logical architecture that integrates all necessary functionalities. The design emphasizes modularity, allowing different components such as appointment scheduling, record management, and prescription handling to function independently while remaining interconnected.

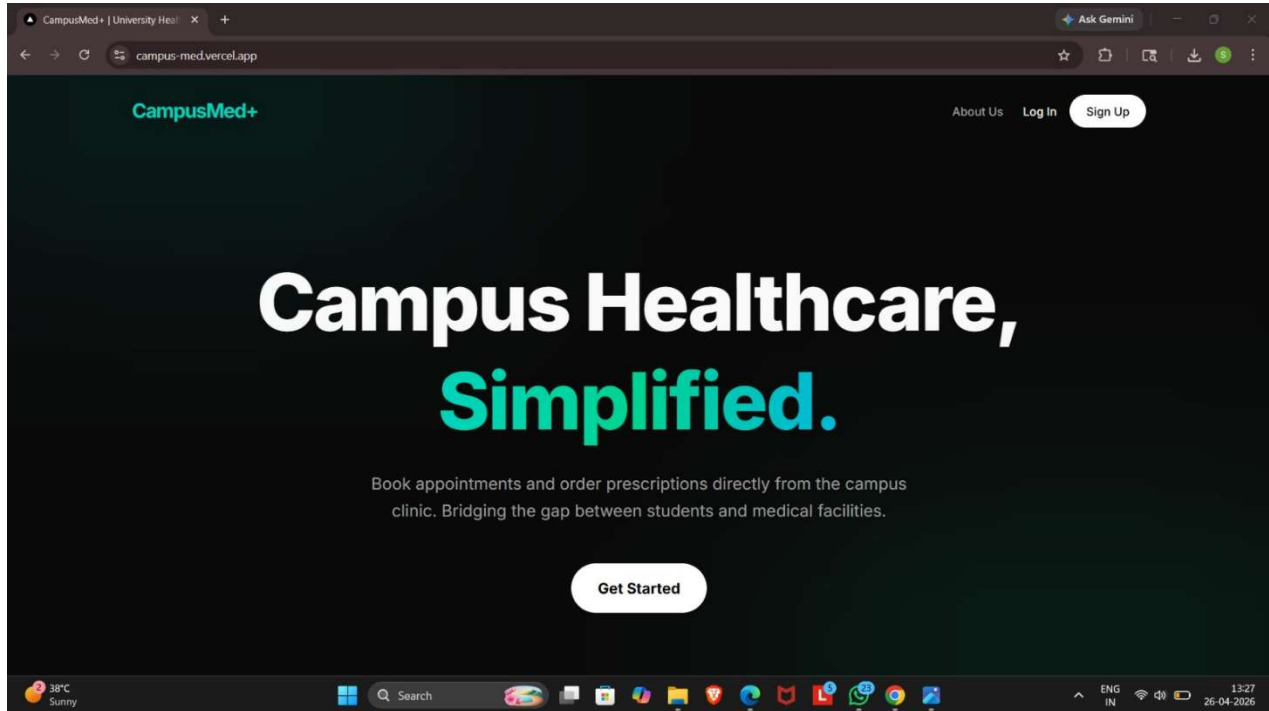
The workflow of the system is carefully structured to ensure smooth interaction between users. The process begins when a student logs into the system using institutional credentials. Once authenticated, the student can view available time slots and book an appointment with a doctor. This information is immediately reflected in the doctor's interface, allowing them to prepare for consultations in advance.

During the consultation, doctors can access the patient's medical history through the system, enabling more informed diagnosis and treatment. After the consultation, digital prescriptions are generated and stored in the database, making them easily accessible for future reference. Medical staff play a supporting role by updating patient records, managing medicine inventory, and coordinating distribution.

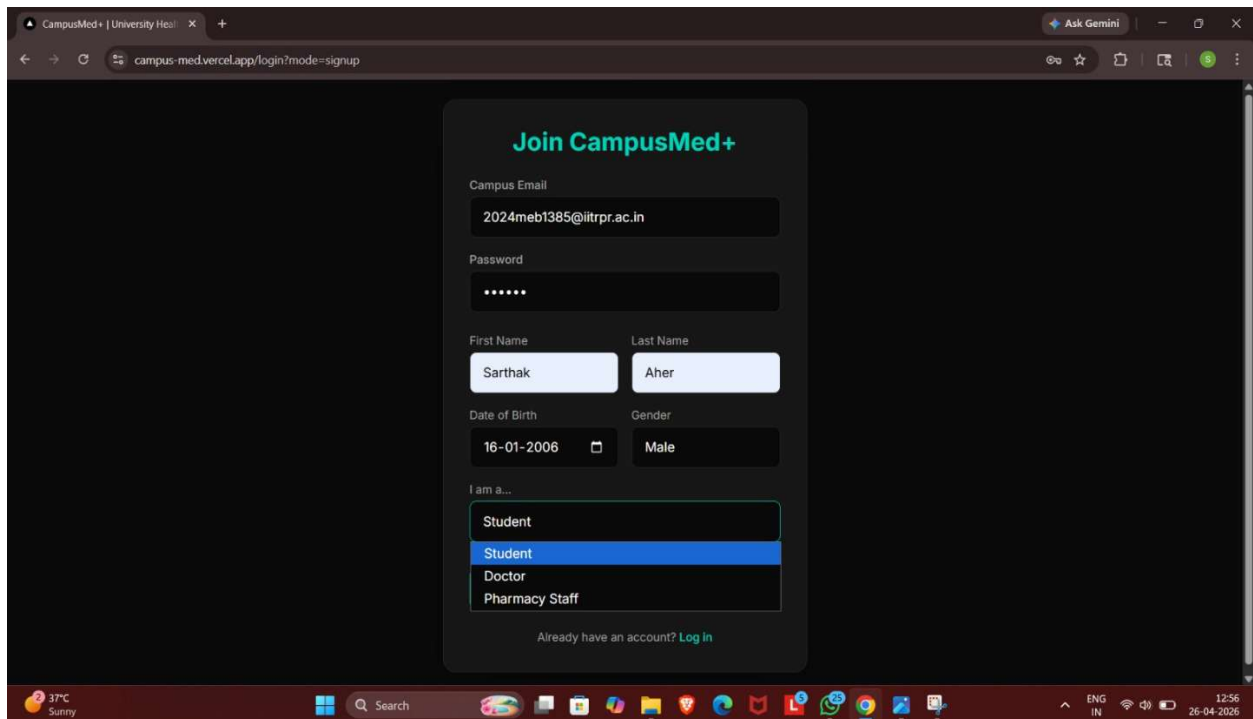
A critical aspect of the system is its backend infrastructure, which is designed to operate within a secure, non-cloud environment. This ensures that all sensitive data remains protected while still allowing efficient data processing and storage. The frontend interface complements this by providing an intuitive and responsive user experience.

Overall, the workplan ensures a systematic approach to development, with each stage building upon the previous one to create a cohesive and functional system. All the things mentioned above were followed to make this website: - <https://campus-med.vercel.app/>

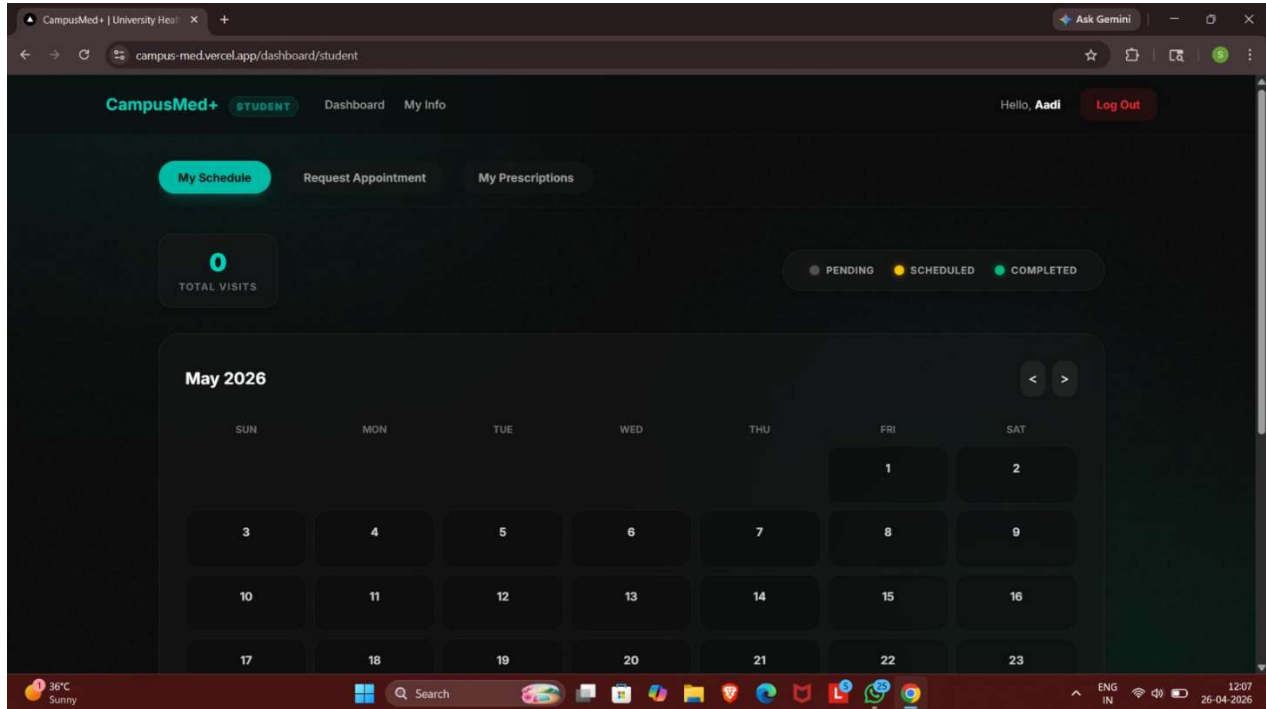
Main Page



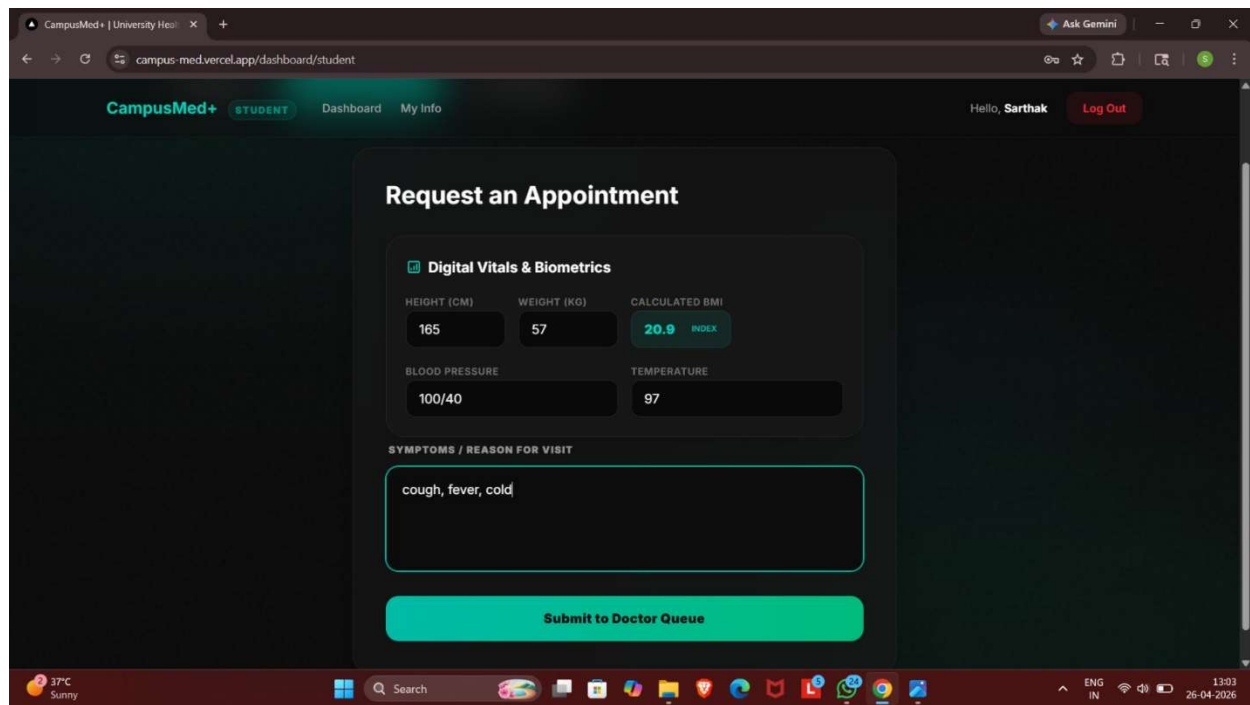
Common Sign-Up Page



Login As Student: My Schedule



Login As Student: Request Appointment



Login As Student: My Prescriptions

The screenshot shows the CampusMed+ Student Dashboard. The top navigation bar includes the CampusMed+ logo, a 'STUDENT' tab, and links to 'Dashboard' and 'My Info'. The user is logged in as 'Aadi' with a 'Log Out' button. The main content area is titled 'Active Prescriptions' with a sub-header 'Action Required'. It displays a prescription for 'RX-9YPSX' with an expiration date of '0202-04-26' by 'Dr. Ranbir Kapoor'. The doctor's notes state 'paracetamol take a good sleep'. Below the notes are three buttons: 'Campus Pharmacy Show code at counter', 'Download PDF Official Prescription', and 'Procured Elsewhere? Mark as Done'. The 'Dispensed History' section shows a table with columns: DATE, CODE, FULFILLMENT, NOTES, and ACTION. The first entry is for '2027-04-05' with code 'RX-YZ3MD', fulfillment 'PHARMACY', and notes '1. paracetmol tab (500 mg) 2. azethromycin tab'. The bottom of the screen shows a Windows taskbar with a search bar and various application icons, and a system tray with weather (36°C Sunny), language (ENG IN), and date/time (12:27, 26-04-2026).

CampusMed+ | University Health | New Tab | Ask Gemini

campus-med.vercel.app/dashboard/student

CampusMed+ STUDENT Dashboard My Info Hello, Aadi Log Out

Active Prescriptions Action Required

RX CODE 0202-04-26
RX-9YPSX Dr. Ranbir Kapoor

DOCTOR'S NOTES / MEDICINES

paracetamol take a good sleep

Campus Pharmacy Show code at counter Download PDF Official Prescription Procured Elsewhere? Mark as Done

Dispensed History

DATE	CODE	FULFILLMENT	NOTES	ACTION
2027-04-05	RX-YZ3MD	PHARMACY	1. paracetmol tab (500 mg) 2. azethromycin tab	PDF

36°C Sunny Search 12:27 26-04-2026

Login As Doctor: FIFO Request Queue

The screenshot shows the CampusMed+ Doctor Dashboard. The top navigation bar is similar to the student view. The main content area is titled 'FIFO Request Queue'. A modal titled 'Assign Time Slot' is open, showing patient details for '2024meb1385@iitrpr.ac.in' with a status of 'AWAITING TRIAGE'. The patient's vitals are displayed: HEIGHT 165 cm, WEIGHT 57 kg, BMI 20.9, BP 100/40, and TEMP 97. The reported symptoms are 'cough, fever, cold'. The modal also includes fields for Date (26-01-2006) and Time (01:30), and buttons for 'Cancel' and 'Confirm Slot'. The bottom of the screen shows a Windows taskbar with a search bar and various application icons, and a system tray with weather (37°C Sunny), language (ENG IN), and date/time (13:04, 26-04-2026).

CampusMed+ | University Health | New Tab | Ask Gemini

campus-med.vercel.app/dashboard/doctor

Assign Time Slot

Patient: 2024meb1385@iitrpr.ac.in AWAITING TRIAGE

HEIGHT 165 cm WEIGHT 57 kg BMI 20.9 BP 100/40 TEMP 97

REPORTED SYMPTOMS

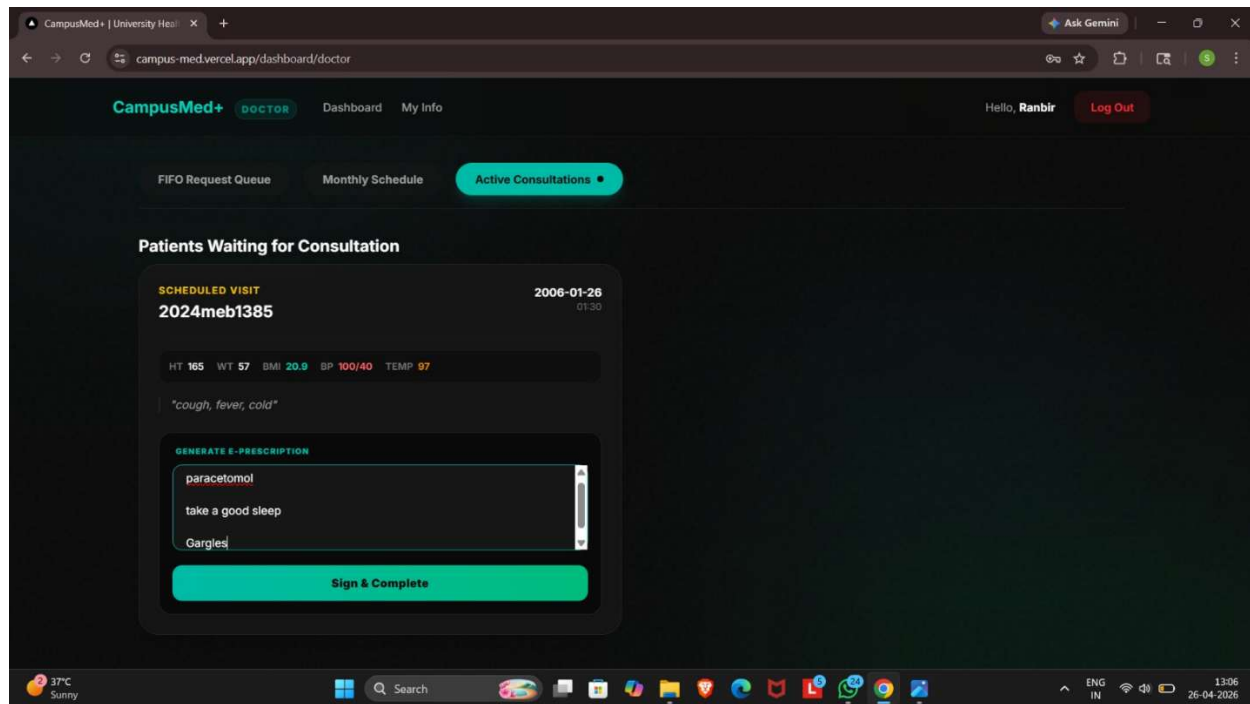
cough, fever, cold

Date 26-01-2006 Time 01:30

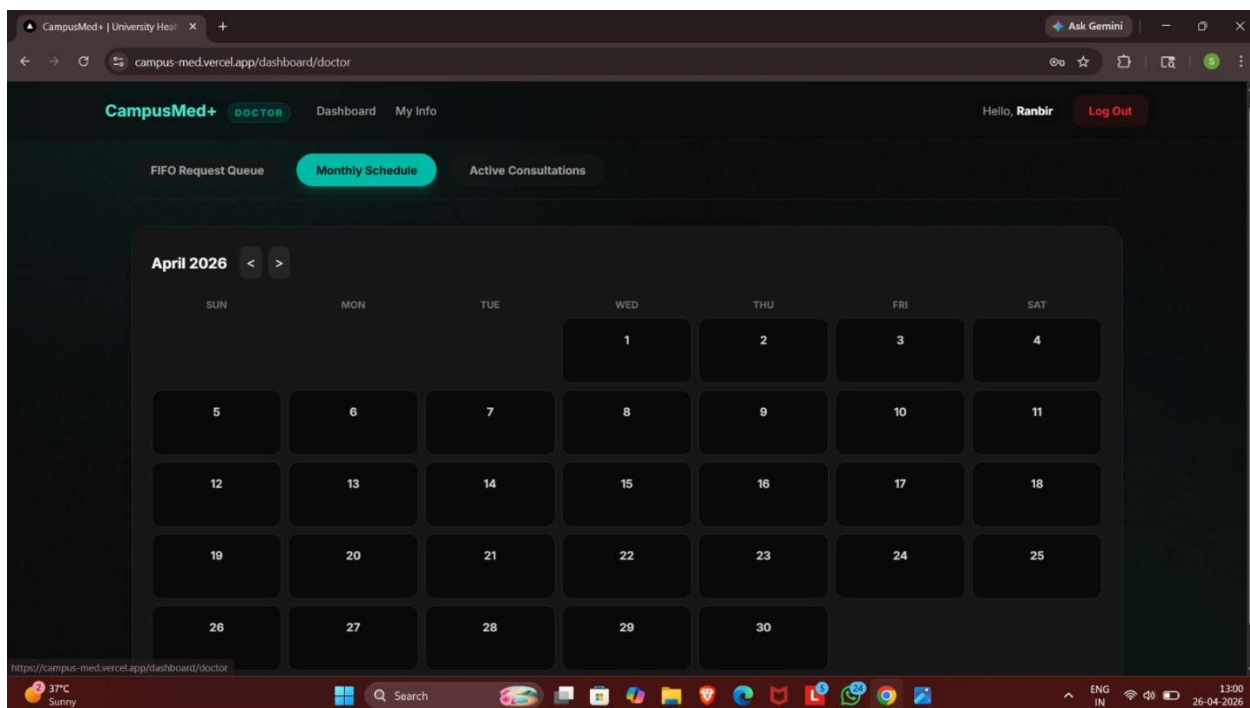
Cancel Confirm Slot

37°C Sunny Search 13:04 26-04-2026

Login As Doctor: Active Consultations/Prescription Writing



Login As Doctor: Monthly Schedule



Login As Pharmacy: Pending Pickups

The screenshot shows the 'Pending Pickups' section of the CampusMed+ Pharmacy Dashboard. The interface includes a header with the CampusMed+ logo, a 'PHARMACY' tab, and navigation links for 'Dashboard' and 'My Info'. The user is logged in as 'Ranveer' and can click 'Log Out'. The main content area has three tabs: 'Pending Pickups' (active), 'Inventory Management', and 'Dispensed History'. Below the tabs is a search bar labeled 'Scan or type RX Code (e.g., RX-9A2B4)...'. A card displays a selected RX code 'RX-YZ3MD' with a date '2024meb1319' and a doctor's name '2027-04-05 • Dr. siddu jnavor'. Under 'DOCTOR'S NOTES', there is a list: '1. paracetmol tab (500 mg) 2. azethromycin tab'. The 'FULFILL ORDER' section shows 'Paracetamol (57 in stock)' with a quantity input of '3' and an 'Add' button. A large green button at the bottom of the card is labeled 'Handover & Mark Dispensed'.

CampusMed+ | University Health

PHARMACY Dashboard My Info Hello, Ranveer Log Out

Pending Pickups Inventory Management Dispensed History

Scan or type RX Code (e.g., RX-9A2B4)...

RX CODE 2024meb1319
2027-04-05 • Dr. siddu jnavor

DOCTOR'S NOTES

1. paracetmol tab (500 mg) 2. azethromycin tab

FULFILL ORDER

Paracetamol (57 in stock) 3 Add

Handover & Mark Dispensed

Login As Pharmacy: Inventory Management

The screenshot shows the 'Inventory Management' section of the CampusMed+ Pharmacy Dashboard. The header and navigation are identical to the previous screen. The 'Inventory Management' tab is active. It features an 'Add New Medicine' form with fields for 'MEDICINE NAME' and 'INITIAL STOCK', and an 'Add to Database' button. Below the form is a table listing current inventory items.

MEDICINE NAME	STATUS	ACTIONS
Dolo	64 in stock	- + Delete
Levocitrizin	58 in stock	- + Delete
Paracetamol	57 in stock	- + Delete
Wow	50 in stock	- + Delete

CampusMed+ | University Health

PHARMACY Dashboard My Info Hello, Ranveer Log Out

Pending Pickups Inventory Management Dispensed History

Add New Medicine

MEDICINE NAME INITIAL STOCK Add to Database

MEDICINE NAME STATUS ACTIONS

Dolo 64 in stock - + Delete

Levocitrizin 58 in stock - + Delete

Paracetamol 57 in stock - + Delete

Wow 50 in stock - + Delete

Login As Pharmacy: Dispensed History

The screenshot shows the 'Dispensed History' section of the CampusMed+ Pharmacy dashboard. The interface includes a header with the CampusMed+ logo, a 'PHARMACY' tab, and navigation links for 'Dashboard' and 'My Info'. The user is logged in as 'Ranveer' and can click 'Log Out'. Below the header, there are three tabs: 'Pending Pickups', 'Inventory Management', and 'Dispensed History' (which is active). The main content area displays a table with the following data:

STATUS	RX CODE	STUDENT	ITEMS DISPENSED
DISPENSED	RX-YZ3MD	2024meb1319	No physical meds given

The bottom of the image shows a Windows taskbar with a search bar, various application icons, and system information indicating 36°C Sunny, 12:15, and 26-04-2026.

Login As Admin: System Overview

The screenshot shows the 'System Overview' section of the CampusMed+ Admin dashboard. The interface includes a header with the CampusMed+ logo, an 'ADMIN' tab, and navigation links for 'Dashboard' and 'My Info'. The user is logged in as 'System' and can click 'Log Out'. Below the header, there are three tabs: 'System Overview' (which is active), 'Pending Approvals', and 'User Directory & Access'. The main content area displays four summary cards:

- 6 TOTAL ACCOUNTS
- 2 STUDENTS
- 0 ACTIVE APPOINTMENTS
- 1 COMPLETED VISITS

Below these cards is a 'Live Appointment Feed' section. It shows a notification for '2024meb1385@litpr.ac.in' assigned to 'Dr. Ranbir Kapoor' on '2006-01-29'. The status is 'COMPLETED'. To the right of the feed, it says 'ADMIN WRITE ACCESS ACTIVE'.

The bottom of the image shows a Windows taskbar with a search bar, various application icons, and system information indicating 38°C Sunny, 13:47, and 26-04-2026.

Login As Admin: Pending Approvals

CampusMed+ADMINDashboardMy InfoHello, SystemLog Out

System OverviewPending Approvals1User Directory & Access

Pending Medical Staff Approvals

Review and verify credentials before granting dashboard access to new staff.

APPLICANT NAME	EMAIL	REQUESTED ROLE	ACTIONS
raj singh	doctor3@iitrpr.ac.in	DOCTOR	RejectApprove Access

38°C Sunny

Search

ENG IN13:5126-04-2026

Login As Admin: User Directory & Access

CampusMed+ADMINDashboardMy InfoHello, SystemLog Out

System OverviewPending ApprovalsUser Directory & Access

System Directory

Warning: Changing roles impacts user access immediately.

NAME & STATUS	EMAIL	SYSTEM ROLE	ACTIONS
System Admin APPROVED	admin@iitrpr.ac.in	ADMIN	Delete Profile
Aadi Shah APPROVED	2024meb1319@iitrpr.ac.in	STUDENT	Delete Profile
gjhkj cbbfh APPROVED	doctor2@iitrpr.ac.in	DOCTOR	Delete Profile
Ranbir Kapoor APPROVED	doctor1@iitrpr.ac.in	DOCTOR	Delete Profile
d 2	d2@iitrpr.ac.in	PHARMACY	Delete Profile

38°C Sunny

Search

ENG IN13:5026-04-2026

Novelty / Innovation of the Proposed Intervention

The proposed system stands out due to its strong focus on context-specific design and practical applicability within an institutional setting. Unlike generic healthcare applications that are designed for large-scale commercial use, this system is tailored specifically to the needs and constraints of the IIT Ropar medical center. This targeted approach ensures that the solution is not only effective but also feasible for implementation.

One of the most significant innovations lies in the emphasis on data privacy. In contrast to most modern healthcare platforms that rely heavily on cloud-based storage, this system is designed to function within a restricted or local network environment. This approach addresses critical concerns related to data security and institutional compliance, making it highly relevant in scenarios where confidentiality is paramount.

Another innovative aspect is the integration of multiple functionalities into a unified platform. By combining appointment scheduling, record management, and prescription handling into a single system, the project eliminates fragmentation and enhances operational efficiency. The system also leverages existing institutional infrastructure, such as student identification systems, to simplify authentication and reduce complexity.

The project's novelty is further enhanced by its origin as a student-driven initiative based on real-world problem identification. This not only demonstrates the practical application of academic knowledge but also highlights the potential for student contributions to institutional development.

Approaches to Implementation

The implementation of the proposed system can be carried out through a phased approach to ensure smooth transition and minimal disruption to existing operations. The first phase involves the development of a prototype that demonstrates core functionalities such as login, appointment booking, and record management. This prototype serves as a testing platform to identify potential issues and gather feedback from users.

Once the prototype is validated, the next phase involves controlled testing within the medical center environment. This allows for real-world evaluation of the system's performance and usability. Feedback collected during this stage can be used to refine the system and address any shortcomings.

The final phase involves full-scale deployment, where the system is integrated into daily operations. This requires coordination with medical staff and administrative authorities to

ensure proper training and adaptation. Continuous monitoring and support are essential during this stage to ensure successful implementation.

Possible Constraints and Barriers

Despite its potential benefits, the project faces several challenges that must be addressed for successful implementation. One of the primary constraints is ensuring data security without relying on cloud infrastructure. Designing a secure local system requires careful planning and may involve additional technical complexity.

Another significant barrier is the limited availability of medical staff. With only a small number of doctors currently serving the campus, the adoption of a digital system may initially increase workload as users adapt to new processes. Resistance to change and lack of familiarity with technology can also pose challenges during implementation.

Administrative approval is another critical factor, as the deployment of such a system requires institutional support and resource allocation. Budget constraints and policy considerations may influence decision-making, potentially delaying implementation.

Expertise Available with Each Student

The successful development of this project is supported by the diverse skill set of the team members. Each student contributes expertise in specific areas, creating a balanced and collaborative working environment. Some members focus on frontend development, designing user interfaces that are intuitive and accessible, while others specialize in backend development, ensuring efficient data management and system functionality.

In addition to technical skills, the team also possesses strengths in research, data analysis, and documentation. These skills are crucial for conducting need assessment, interpreting stakeholder feedback, and presenting findings in a structured manner. The combination of technical and analytical expertise enables the team to approach the project holistically, addressing both design and implementation challenges effectively.

Expected Outcomes

The project is expected to result in the development of a functional prototype that demonstrates the feasibility and effectiveness of a digital healthcare management system

within the IIT Ropar medical center. This prototype will showcase improvements in key areas such as record management, appointment scheduling, and overall service efficiency.

By reducing reliance on manual processes, the system is anticipated to minimize errors, improve data accessibility, and enhance the quality of healthcare services. Students will benefit from faster and more convenient access to medical assistance, while doctors and staff will experience improved workflow management.

In the long term, the project has the potential to serve as a scalable model that can be adapted for use in other institutions. It also contributes to the broader goal of integrating technology into healthcare systems to improve outcomes and efficiency.

Suggested Plan of Action for Utilization of Outcomes

The outcomes of this project can be effectively utilized by presenting the developed prototype to the institute administration as a proof-of-concept. This presentation should highlight the system's features, benefits, and alignment with institutional needs, thereby facilitating informed decision-making.

Based on administrative feedback, further development can be undertaken to enhance the system and address any identified gaps. Collaboration with technical and medical experts may be required to ensure that the system meets all operational and regulatory requirements.

Once approved, the system can be gradually integrated into the medical center's workflow, starting with limited functionality and expanding over time. Continuous evaluation and improvement are essential to ensure long-term success and sustainability.

Conclusion

This project demonstrates the importance of adopting a need-based approach in addressing real-world challenges. By focusing on the specific requirements of the IIT Ropar medical center, it proposes a solution that is both practical and impactful. The integration of technology into healthcare management not only improves efficiency but also enhances the overall experience for users.

The project also highlights the value of stakeholder engagement in designing effective solutions. By incorporating insights from medical staff and administrative authorities, the

proposed system aligns closely with actual needs and constraints. This ensures that the solution is not only innovative but also implementable.

Ultimately, the project represents a step towards modernizing institutional healthcare systems, showcasing how student-driven initiatives can contribute meaningfully to societal and organizational development.